功能

DILIrank 2.0：基于 FDA 标签和文献综述的药物性肝损伤风险更新和扩展数据库

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药物性肝损伤（DILI）在药物开发和公共卫生中备受关注。DILIrank 1.0 是一个广泛使用的公共数据集，根据药物引起 DILI 的潜在可能性对 FDA 批准的药物进行排名，这显著促进了改进 DILI 评估的新方法的发展。在此，我们介绍 DILIrank 2.0，这是 DILIrank 1.0 的重要更新，用于捕捉自其 15 年前推出以来产生的新非生物药物和与肝脏相关的不良事件数据。利用 DILIrank 2.0，我们还观察到在引入不同 FDA 监管项目后，获批药物的 DILI 特征发生了变化。DILIrank 2.0 是一个最新的资源，为支持 DILI 安全性评估、预测模型开发以及新方法（NAMs）的评估提供了机会。

关键词：药物性肝损伤（DILI）；DILIrank；肝毒性；数据库；药物安全

Feature • PERSPECTIVE

介绍

DILI 在 20 世纪末成为医疗保健中的一个关键问题，这在很大程度上归功于 Hyman Zimmerman 的开创性工作。自那时以来，对 DILI 的研究兴趣显著增加。研究内容包括但不限于：表征 DILI 的临床病例，根据药物的肝毒性潜力对药物进行分类，识别与药物相关和宿主相关的风险因素，标准化诊断标准，阐明其潜在机制肝毒性机制，(p6)，(p7) 以及开发用于DILI评估的预测模型。(p8)，(p9) 这些努力旨在增强对DILI发病机制的理解，改善早期检测和预防策略，并最终降低其在人类中的发生率。该领域的进展有望减轻与DILI相关的医疗负担，并通过减少由肝毒性造成的晚期阶段流失，提高药物开发的成功率。

根据药物在人类中引起DILI的风险对其进行标注，是推进DILI研究的关键步骤，尤其是在开发预测模型方面，这些模型利用现有的肝毒性潜力知识来构建用于模型开发的数据集。以前，我们引入了一个DILI基准数据集，这是一个基于从FDA批准的药物标签中提取的数据的系统性DILI分类方案。(p10)随后，通过整合，进一步完善了这一分类。

[1] 这些作者对本研究贡献相同。

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Feature

DILIrank 2.0: An updated and expanded database for drug-induced liver injury risk based on FDA labeling and a literature review

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Drug-induced liver injury (DILI) is of great concern in drug development and public health. DILIrank 1.0, a widely used public dataset that ranks FDA-approved drugs by their potential to cause DILI, has significantly enabled the development of new methods for improved DILI assessment. Here, we introduce DILIrank 2.0, an essential update of DILIrank 1.0, to capture new non-biologics drug and liver-related adverse event data generated since its inception 15 years ago. Using DILIrank 2.0, we also observe changes in the DILI profiles of approved drugs following the introduction of different FDA regulatory programs. DILIrank 2.0 is an up-to-date resource that offers opportunities for supporting DILI safety assessment, predictive model development, and evaluation of new approach methods (NAMs).

Keywords: drug-induced liver injury (DILI); DILIrank; hepatotoxicity; database; drug safety

Feature • PERSPECTIVE

Introduction

DILI emerged as a critical concern in healthcare during the late 20th century, largely because of the seminal work of Hyman Zimmerman.(p1) Since then, research interest in DILI has expanded significantly. Investigations have included, but are not limited to, characterizing clinical cases of DILI, categorizing drugs based on their hepatotoxic potential,(p2) identifying drug-related and host-related risk factors,(p3),(p4) standardizing diagnostic criteria,(p5) elucidating the underlying mechanisms of hepatotoxicity,(p6),(p7) and developing predictive models for DILI assessment.(p8),(p9) These efforts aim to enhance the understanding of DILI pathogenesis, improve early detection and prevention strategies, and ultimately reduce its incidence in humans. Advancements in this field are expected to alleviate the healthcare burden associated with DILI and improve the success rates of drug development by reducing late-stage attrition caused by hepatotoxicity.

Annotating drugs based on their DILI risk in humans is a critical step toward advancing the study of DILI, and especially in developing predictive models, which utilize existing knowledge of hepatotoxic potential to construct datasets for model development. Previously, we introduced a DILI benchmark dataset, a systematic classification scheme for DILI based on data extracted from the FDA-approved drug labels.(p10) This classification was later refined by incorporating

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临床病例数据，其中DILI因果关系通过标准化评估方法验证，如Roussel Uclaf因果关系评估方法（RUCAM）或专家共识。[p11] 由此产生的DILIrank（版本1.0）数据集，包含1036种药物，其影响显著，表现为在科学界的广泛认可。DILIrank 1.0已被推荐用于DILI注释[p12],[p13]，并且在众多研究中作为DILI分类的参考，包括计算机模型开发、评估和验证研究[p14],[p15],[p16],[p17],[p18],[p19]；用于DILI风险因素的识别[p20],[p21]；以及用于评估体外试验预测性能。[p9],[p22]

对DILI特定药物数据库的全面回顾显示，DILIrank 1.0是一个有价值的资源，其显著的限制是需要更新最近由FDA批准的新药分子。DILIrank 1.0仅包含截至2009年由FDA批准的药物。鉴于FDA每年批准约40种新药，因此2010年至2021年间批准的约400种药物未包含在DILIrank 1.0中。此外，一些15年前被注释的药物也出现了新的安全性问题。为了确保DILIrank数据集保持最新状态并包含最近FDA批准的药物，需要对DILIrank数据集进行新的更新。

在这里，我们介绍 DILIrank 2.0，这是DILIrank 数据库急需的更新版。DILIrank 2.0 融入了先前版本的修订内容，这些修订基于文献中新报告的 DILI 病例，并对 FDA 2021 年之前批准的额外药物进行了注释，以认识到在上市后监测期间发现和报告安全问题所需的时间。除了更新数据库之外，我们还分析了历史监管政策对 FDA 批准药物中 DILI 情况的影响。

DILIrank 2.0 第I部分 – DILIrank 1.0 药物注释的更新

鉴于人类中药物性肝损伤（DILI）的发生率较低，以及现有预测模型在准确识别肝毒性方面的局限性，一些有毒药物可能在临床前测试和临床试验中逃避检测。这一挑战尤其特别是对于那些以特异性方式引起肝损伤的药物。因此，上市后监测需要足够的时间，以便开发和评估临床病例报告，从而确认药物性肝损伤（DILI）的因果关系。

在这里，我们通过整合文献中新报告的DILI病例，重新评估了DILIrank 1.0的注释，重点关注之前被归类为^{v}无DILI关注或DILI关注不明确的药物。相比之下，之前归入 ^{v}低DILI关注和 ^{v}高DILI关注组的药物已经符合我们的验证标准，其DILI分类保持不变（图1a）。

为了评估近期有关药物性肝损伤（DILI）因果关系的证据，我们进行了文献检索，检索对象为同行评审的病例报道，使用的关键词为: [药物名称] AND（肝损伤 OR DILI OR 肝毒性 OR RUCAM OR CIOMS OR Naranjo OR 因果关系 OR 诱发）。仅考虑用英文撰写且病例通过因果关系评估方法（CAM）如 RUCAM 进行裁定的报告。与 DILIrank 1.0 类似，只有当 CAM 评分在 Naranjo 量表中被归类为“确定”或“可能”，或在国际医学科学组织委员会（CIOMS）/RUCAM 量表中归类为“高度可能”、“可能”或“可能性存在”时，药物才被“验证”为导致 DILI。若未找到病例报告，或现有病例报告未应用 CAM 评分，或 CAM 评分适用于两种或以上药物的组合，则该药物被认为“未验证”为 DILI 的原因。在线补充材料表 S1 列出了这些具有 DILI 相关病例报告的药物。

在被归类为 ^{v} 无 DILI 关注、DILIrank 1.0 的 312 种药物中，有 43 种药物发现了 DILI 病例报告。其中，18 种药物符合我们因果关系验证标准，并在 DILIrank 2.0 中重新归类为 ^{v} 较少 DILI 关注，而其余 25 种药物仍保留其 ^{v} 无 DILI 关注分类。类似地，在之前被归类为 模糊 DILI 关注 的 254 种药物中，有 76 种药物发现了 DILI 病例报告。其中，31 种药物的新验证显示可引起 DILI。根据药物标签信息及其描述的 DILI 严重程度根据FDA批准的药物标签，27种药物被重新分类为 ^{v}低肝损伤风险，而4种药物被重新分类为 ^{v}高肝损伤风险。其余45种药物的肝损伤分类保持不变（图1a）。

DILIrank 2.0 第二部分 – 新增 300 种药物

由于 DILIrank 1.0 包含的药物仅批准至 2009 年，因此有必要对自那以后获批的新药进行彻底检查和注释。因此，除了更新 DILIrank 1.0 外，我们还扩展了数据库，通过应用我们以前用于 DILIrank 1.0 的模式（如 Chen 等人所述[p11] （图 1b））来分类 2010 年至 2021 年间 FDA 批准的药物。选择过程涉及根据以下标准筛选药物:

(i) FDA 批准药物的新药申请 (NDA) 类型：1 型 NDA（新分子实体）或 2 型 NDA（非新分子实体的新活性成分）；(ii) 仅含单一活性成分的药物；(iii) 仅用于人体的药物；(iv) 仅通过口服或注射途径给药的药物；以及 (v) 拥有有效 FDA 药物标签的药物。

对于符合这些标准的药物，从 FDALabel数据库[p24]（版本2.7.1，2023年2月16日）中检索了药物标签的最新版本，并用于分配DILI严重性并建立初始DILI分类，遵循先前描述的方法论。[p10]

最后，使用与前一节中概述的 DILI 注释重新评估中所采用的相同方法，验证了分配的 DILI 分类。对于在 FDA 标签文件的框中警告部分中描述了肝毒性问题的药物，被认为已被验证为会引起 DILI，如前所述。

共有300种新批准的FDA药物被纳入 DILIrank 2.0。在这些药物中，49种药物被证实会引起DILI，其中28种药物被归类为^{v}较低DILI关注，21种被归类为 ^{v}高度DILI关注。此外，有120种药物被分类为-

clinical case data, where DILI causality was verified through standardized assessment methods, such as the Roussel Uclaf Causality Assessment Method (RUCAM), or expert consensus.[p11] The resulting DILIrank (version 1.0) dataset, comprising 1036 drugs, has had a significant impact, as evidenced by its extensive acceptance within the scientific community. DILIrank 1.0 has been recommended for DILI annotation use[p12],[p13] and has served as a reference for DILI classification in numerous studies, including studies on *in silico* model development, assessment, and validation[p14],[p15],[p16],[p17],[p18],[p19]; for identification of DILI risk factors[p20],[p21]; and in the evaluation of *in vitro* assay predictive performance.[p9],[p22]

A comprehensive review of DILI-specific drug databases highlighted DILIrank 1.0 as a valuable resource, with a notable limitation being the need for an update of new drug molecules approved by the FDA recently.[p23] DILIrank 1.0 contains drugs approved by the FDA only up to 2009. Given that the FDA approves ~40 new drugs annually, ~400 drugs approved between 2010 and 2021 are not present in DILIrank 1.0. Additionally, new safety concerns have emerged for some drugs that were annotated 15 years ago. To ensure that the DILIrank dataset remains current and includes the most recent FDA-approved drugs, a new update of DILIrank dataset is required.

Here, we present DILIrank 2.0, a critically needed update to the DILIrank database.[p11] DILIrank 2.0 incorporates revisions of the previous version based on DILI cases newly reported in the literature and annotating additional drugs approved by the FDA through 2021, in recognition of the need for time to discover and report safety concerns during post-marketing surveillance. In addition to updating the database, we also analyze the impact of historical regulatory policies on the DILI landscape among FDA-approved drugs.

DILIrank 2.0 Part I – update to DILIrank 1.0 drug annotations

Given the low incidence of DILI in humans and the current limitations of predictive models in accurately identifying hepatotoxicity, some toxic drugs might evade detection during preclinical testing and clinical trials. This challenge is particularly pronounced for drugs that induce liver injury in an idiosyncratic manner. Therefore sufficient time is required for post-marketing surveillance to allow for the development and assessment of clinical case reports confirming DILI causality.

Here, we reassessed the annotations from DILIrank 1.0 by incorporating newly reported DILI cases from the literature, focusing on drugs previously classified as ^{v}No-DILI-Concern or Ambiguous-DILI-Concern. In contrast, drugs previously categorized under the ^{v}Less- and ^{v}Most-DILI-Concern groups already met our verification criteria, and their DILI classification remained unchanged (Figure 1a).

To evaluate recent evidence of DILI causality, we conducted a literature search for peer-reviewed case reports using the keywords: [drug name] AND (liver injury OR DILI OR hepatotoxicity OR RUCAM OR CIOMS OR Naranjo OR causality OR induced). Reports were considered only if they were written in English and the cases were adjudicated using a causality assessment method (CAM) such as RUCAM. Similar to DILIrank 1.0, a drug was only 'verified' as causing DILI if the CAM score was categorized as 'definite' or 'probable' in the Naranjo scale; or 'highly probable', 'probable', or 'possible' in the Council for International Organizations of Medical Sciences (CIOMS)/RUCAM scales. A drug was deemed 'not verified' as a cause of DILI if no case reports were found, if the available case reports did not apply a CAM score, or if the CAM score was assigned to a combination of two or more drugs. Table S1 in the supplementary material online lists these drugs, which had DILI-related case reports but did not meet our verification criteria, along with the references to the relevant articles.

Of the 312 drugs classified as ^{v}No-DILI-Concern in DILIrank 1.0, DILI case reports were found for 43 drugs. Among these, 18 drugs met our verification criteria for causality and were reclassified as ^{v}Less-DILI-Concern in DILIrank 2.0, whereas the remaining 25 drugs retained their ^{v}No-DILI-Concern classification. Similarly, DILI case reports were found for 76 out of 254 drugs previously categorized as Ambiguous-DILI-Concern. Of these, 31 drugs were newly verified to cause DILI. Based on the drug labeling information and severity of DILI described in their FDA-approved drug labels, 27 drugs were reclassified as ^{v}Less-DILI-Concern, whereas 4 drugs were reclassified as ^{v}Most-DILI-Concern. The DILI classification for the remaining 45 drugs was left unchanged (Figure 1a).

DILIrank 2.0 Part II – addition of 300 new drugs

Because DILIrank 1.0 contains drugs approved up to 2009, there is a need to thoroughly examine and annotate new drugs that have been approved since then. Thus, in addition to updating DILIrank 1.0, we also expanded the database to classify drugs approved by the FDA from 2010 to 2021 by applying our former schema, as used for DILIrank 1.0 and described in Chen *et al.*[p11] (Figure 1b). The selection process involved filtering drugs based on the following criteria:

(i) type of new drug application (NDA) for FDA-approved drugs: type 1 NDA (new molecular entity) or type 2 NDA (new active ingredient that is not a new molecular entity);
(ii) drugs containing a single active ingredient only;
(iii) drugs intended for human use only;
(iv) drugs administered exclusively via oral or parenteral routes; and
(v) drugs possessing an active FDA drug label.

For drugs meeting these criteria, the most recent version of the drug label was retrieved from the FDALabel database[p24] (version 2.7.1, 16 February 2023) and used to assign DILI severity and establish the initial DILI classification, following the previously described methodology.[p10] Finally, the assigned DILI classification was verified using the same approach applied in the re-evaluation of DILI annotation, as outlined in the preceding section. Drugs for which hepatotoxicity concerns were described in boxed warning sections of the FDALabel document were considered verified as causing DILI, as previously done.

A total of 300 new FDA-approved drugs were incorporated into DILIrank 2.0. Among these, 49 drugs were verified as causing DILI, with 28 drugs classified as ^{v}Less-DILI-Concern and 21 as ^{v}Most-DILI-Concern. Additionally, 120 drugs were cat-

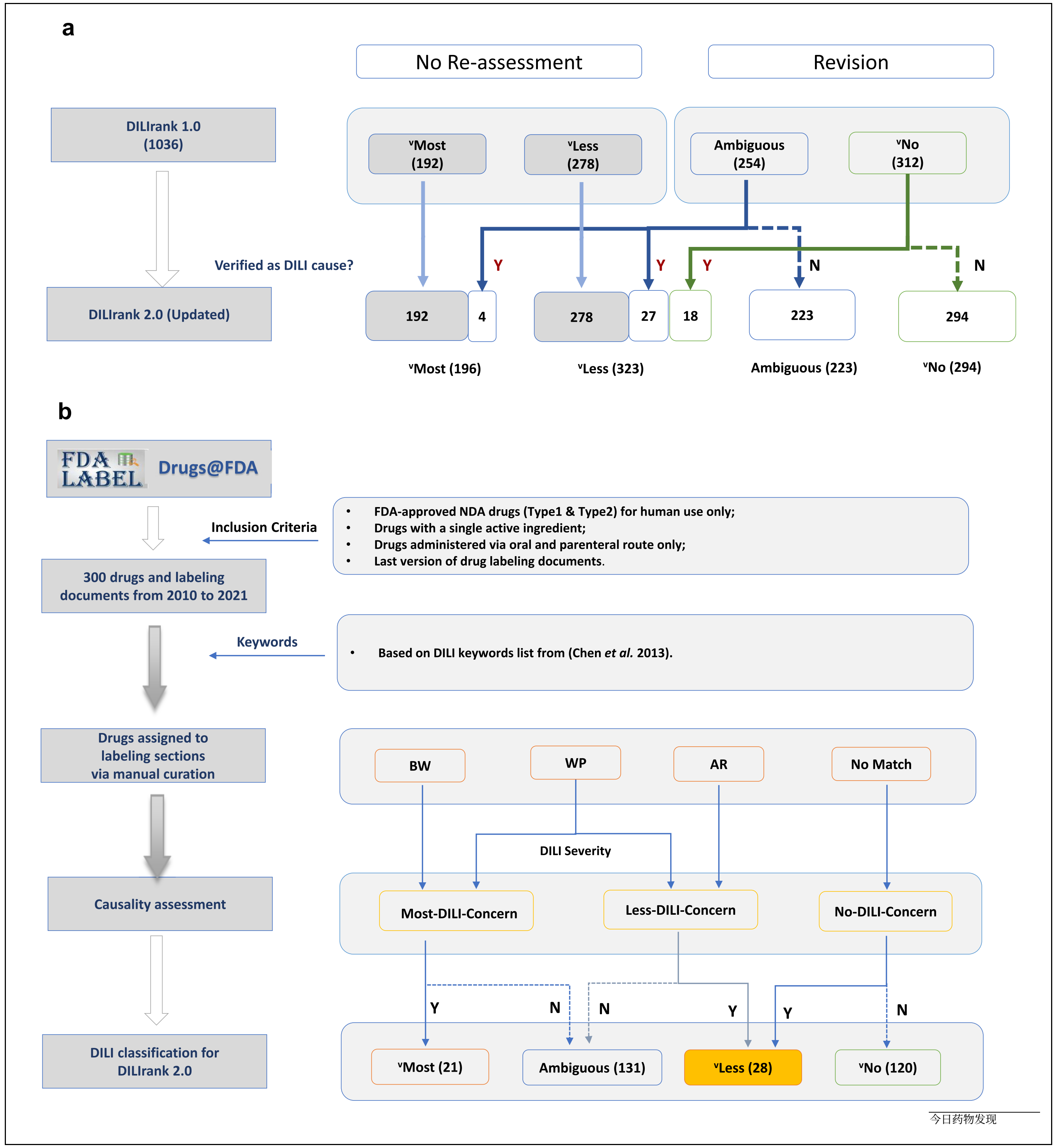

图 1
DILIrank 2.0 注释工作流程。根据可用文献，药物要么被确认会引起 DILI（Y），要么不会（N）。DILI 分组为 ^{v}高度 DILI 风险（^{v}Most）、模糊 DILI 风险（Ambiguous）、^{v}低 DILI 风险（^{v}Less）或 ^{v}无 DILI 风险（^{v}No）。(a) DILIrank 1.0 中 49 种药物的 DILI 注释修订。(b) 2010 至 2021 年 FDA 批准的 300 种药物的 DILI 分类与注释。包含与我们关键词匹配的 DILI 信息的药物标签部分为盒装警告（BW）、警告与注意事项（WP）以及不良反应（AR）；如果在标签文件的这些部分没有信息，则使用“无匹配”（参见 Chen 等人 2016[p11]）。

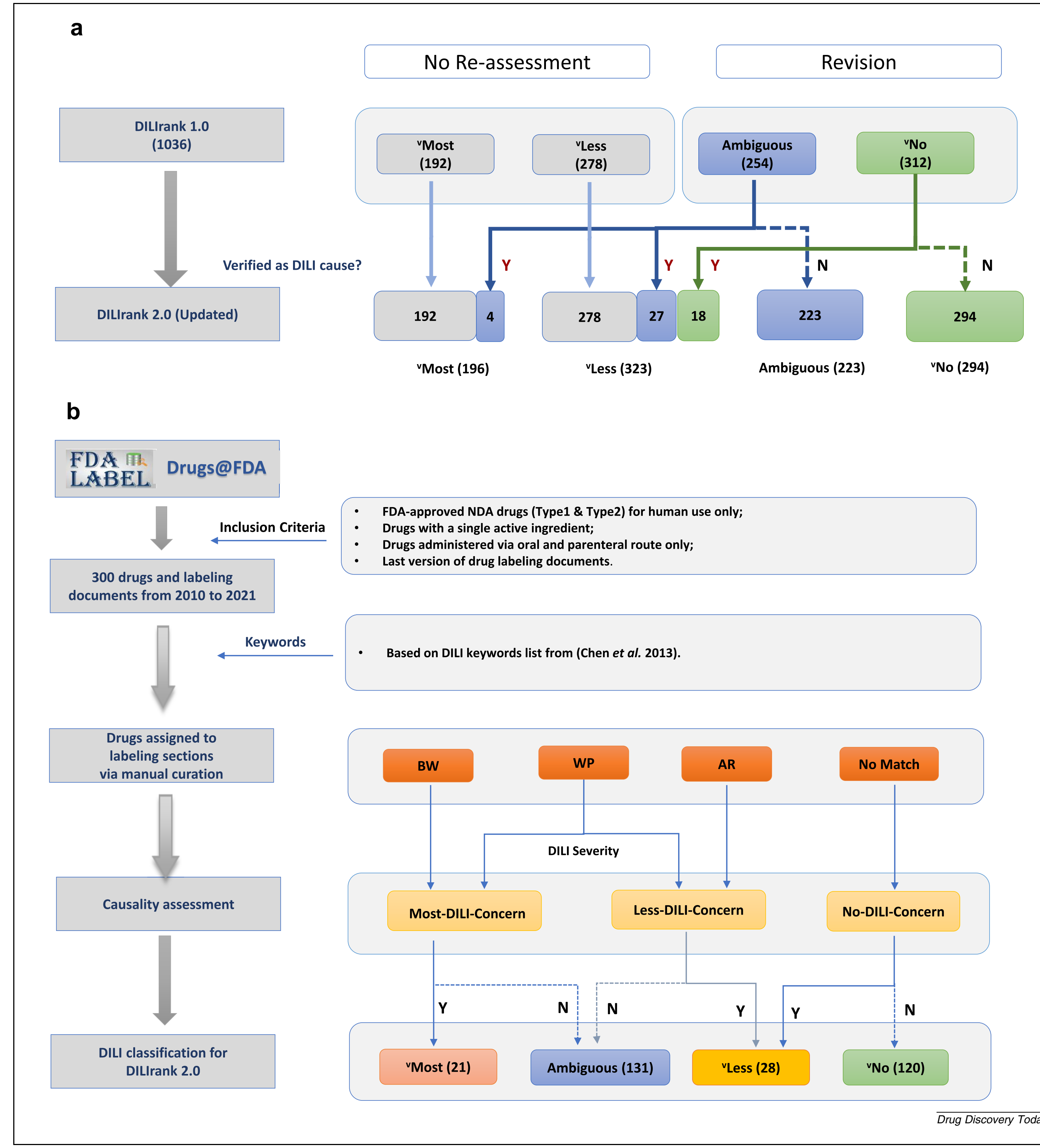

FIGURE 1
Workflows for the annotation of DILIrank 2.0. Based on available literature, a drug was either verified to cause DILI (Y) or not (N). DILI is grouped as ^{v}Most-DILI-Concern (^{v}Most), Ambiguous DILI-Concern (Ambiguous), ^{v}Less-DILI-Concern (^{v}Less), or ^{v}No-DILI-Concern (^{v}No). (a) Revision of the DILI annotation of 49 drugs in DILIrank 1.0. (b) DILI classification and annotation for 300 drugs approved by the FDA from 2010 to 2021. Drug labeling sections containing DILI information matching our keywords were boxed warning (BW), warnings and precautions (WP), and adverse reactions (AR); 'No Match' was used if no information was in these sections of the labeling document (See Chen et al. 2016[p11]).

被归类为 v无肝损伤关注，而其余的131种药物被归类为肝损伤关注不明确。

DILIrank 2.0 现状分析

经过修订和扩展，DILIr-ank 2.0 共包含 1336 种药物，分类如下：414 v无 DILI 关注，354 种模糊 DILI 关注，351 v低 DILI 关注，以及 217 v高度 DILI 关注药物。随着在 DILIrank 1.0 中增加 300 种药物，DILIrank 2.0 的药物总数增加了 29%（表 1）。尽管如此，有趣的是，在 DILIrank 2.0 中，被归类为 v高度 DILI 关注的药物数量比 DILIrank 1.0 减少了 3%。同时，v无 DILI 和模糊 DILI 关注的药物（尚未确认会导致 DILI 的药物）在 DILIrank 2.0 中所占比例增加。

在 DILIrank 2.0 中，不包含 2010 年至 2021 年间经 FDA 批准的任何生物制品，因为这些产品需要通过生物制品许可申请（BLA）获得批准，这一过程与 NDA 途径不同。此外，对于根据 2 型 NDA 批准的药物，同一基础药物的不同盐形式在 DILIrank 2.0 中被作为单独条目列出（例如 amifampridine 和 amifampridine 磷酸盐）。这种区别承认不同盐形式可能表现出独特的药代动力学特性，包括肝脏暴露，这可能是 DILI 的关键因素。此外，这种差异也反映在药物说明书中。例如，amifampridine 的药物说明书不包含任何肝毒性相关信息，因此在 DILIrank 2.0 中被归类为 v 无 DILI 风险，其 DILI 严重性为 0。相反，药物的不良反应部分则...

关于磷酸阿米法普啶的标签报告在临床试验中观察到肝酶升高，导致其被归类为“DILI关注不明确”，DILI严重性为3。在本研究进行时，这两种药物都未达到确认DILI因果关系的验证标准。

为了评估两版 DILIrank 中药物的化学空间分布，使用 Python 中的 RDKit 模块，从 PubChem 数据库获取化学结构的 1265 种药物计算了 MACCS 分子指纹。[p25],[p26] 分子指纹经主成分分析后，前三个主成分解释了总变异的 23%（见在线补充材料中的图 S1）。因此，使用训练于分子指纹主成分上的 t 分布随机邻域嵌入（SNE）模型对化学空间进行可视化（图 2a）。DILIrank 2.0 中药物的化学空间覆盖范围相比 DILIrank 1.0 略有扩展，这表明与 DILIrank 1.0 中的药物相比，DILIrank 2.0 新增药物占据的化学空间没有显著差异。

此外，我们确定了五对具有“活性悬崖”的新药对，其中至少有一种药物是 DILIrank 2.0 的新加入（表 2）。这些药物对具有相似的化学结构和治疗特征，但在人类中具有明显不同的肝毒性，从而补充了先前发表的列表。药物的治疗特征是使用 WHO 解剖学治疗化学（ATC）分类系统确定的。该系统为药物提供多级分类。

解剖/药理、治疗和化学（ATC）分类，并被世卫组织认可为药物使用监测和研究的国际标准。在本研究中，药物对的选择使用了以下标准：

(i) 化学相似性：基于（分子访问系统）（MACCS）分子指纹计算的Tanimoto相似系数至少为0.6。

(ii) 治疗相似性：该药物对属于世界卫生组织ATC分类系统的第4级ATC亚组，但不包括以“其他”开头的非特定亚组。

(iii) 肝毒性差异：该对药物中一种被归类为 v无肝损伤关注，另一种被归类为v最高肝损伤关注（DILIrank 2.0）。

DILIrank 2.0 包含新增到所有 14 个一级 WHO ATC 分类类别中的新药。一些药物有多个 ATC 代码，并且被归类到所有相关的治疗类别中。大多数 ATC 类别的相对规模在 DILIrank 1.0 和 DILIrank 2.0 之间相当，唯一的主要例外是 L 类（抗肿瘤和免疫调节剂），该类别的药物数量从 144 种（12%）增加到 244 种（16%）（图 2b）。这一显著增加反映了 FDA 在过去十年批准了 100 种与癌症相关的药物。

根据世界卫生组织ATC分类系统的第二级（ATC 2级），该级别描述了药物的药理学或治疗作用，在DILIrank 2.0中新增了78种抗肿瘤药（L01）和17种免疫抑制剂（L04）。所有DILIrank 2.0药物的ATC 2级子类名称列在在线补充材料的表S2中。大多数新添加的、经过验证会导致DILI的药物（分类为v低DILI关注或 v高DILI关注）属于这两个类别。按治疗作用和DILI分类对DILIrank 2.0药物分布的分析显示，L01子类具有最多的DILI阳性（v低DILI关注和 v高DILI关注）药物（图 3）。富集DILI阳性药物的子类已标记

表格 1

DILIrank 1.0 和 DILIrank 2.0 之间的差异。

	DILI排名 1.0[a]	DILI等级 2.0[a]	增加[b]
药物总数	1036	1336	300（29%）
每个 DILI 组的药物数量 (%)			
v无DILI顾虑	312（30%）	414（31%）	102（33%）
模糊的药物性肝损伤担忧	254（25%）	354（26%）	100（39%）
v 更少肝损伤担忧	278（27%）	351（26%）	73（26%）
v最具DILI关注	192（19%）	217（16%）	25（13%）

a 基于数据库版本中所有组计算的百分比。b 基于两个数据库版本中每个组的增加计算的百分比。

egorized as vNo-DILI-Concern, whereas the remaining 131 drugs were designated as Ambiguous-DILI-Concern.

Analysis of the DILIrank 2.0 landscape

Following revision and expansion, DILIrank 2.0 contains a total of 1336 drugs, categorized as follows: 414 vNo-DILI-Concern, 354 Ambiguous DILI-Concern, 351 vLess-DILI-Concern, and 217 vMost-DILI-Concern drugs. With the addition of 300 drugs to DILIrank 1.0, the total number of drugs in DILIrank 2.0 increased by 29% (Table 1). Despite this, it is interesting to note that in DILIrank 2.0, the number of drugs classified as vMost-DILI-Concern decreased by 3% compared with DILIrank 1.0. Meanwhile, the vNo- and Ambiguous-DILI-Concern drugs (drugs that were not verified to cause DILI) increased in proportion in DILIrank 2.0.

No biological product approved by the FDA between 2010 and 2021 is included in DILIrank 2.0, as these require approval through a Biologics License Application (BLA), a process distinct from the NDA pathway. Additionally, for drugs approved under a type 2 NDA, different salt forms of the same base drug were included as separate entries in DILIrank 2.0 (e.g. amifampridine and amifampridine phosphate). This distinction acknowledges that different salt forms can exhibit unique pharmacokinetic properties, including hepatic exposure, which could be a critical factor in DILI. Furthermore, this disparity was also reflected in the drug labels. For instance, the drug label for amifampridine does not contain any hepatotoxicity-related information, leading to its classification as vNo-DILI-Concern in DILIrank 2.0, with a DILI severity of 0. In contrast, the adverse reactions section of the drug label for amifampridine phosphate reports elevations in liver enzymes observed in clinical trials, resulting in its classification as Ambiguous-DILI-Concern, with a DILI severity of 3. At the time of this study, neither drug met the verification criteria for confirmed DILI causality.

To assess the chemical space distribution of drugs in both versions of DILIrank, the MACCS molecular fingerprint was computed for 1265 drugs whose chemical structures were retrieved from the PubChem database using the RDKit module in Python.[p25],[p26] The first three principal components, following a principal component analysis of the molecular fingerprints, account for 23% of the total variation (see Figure S1 in the supplementary material online). Thus, a visualization of the chemical space is given using a t-distributed stochastic neighbor embedding (SNE) model trained on principal components of the molecular fingerprints (Figure 2a). The chemical space coverage of drugs in DILIrank 2.0 reveals a slight expansion from that of DILIrank 1.0, suggesting that there is no significant difference in the chemical space occupied by the newly added drugs in DILIrank 2.0 compared with drugs in the DILIrank 1.0.

In addition, we identified five new pairs of drugs with ‘activity cliffs’, where at least one drug of the pair is a new addition to DILIrank 2.0 (Table 2). These drug pairs have similar chemical structures and therapeutic profiles but have significantly distinct hepatotoxicity in humans, supplementing a previously published list.[p8] The therapeutic profiles of drugs were determined using the WHO Anatomical Therapeutic Chemical (ATC) classification system. The system provides multilevel classification for drugs into anatomical/pharmacological, therapeutic, and chemical (ATC) groups, and is endorsed by the WHO as an international standard for drug utilization monitoring and research.[p27],[p28] In this study, the drug pairs were selected using the following criteria:

(i) Chemical similarity: the Tanimoto similarity coefficient, calculated based on the (Molecular ACCess System) (MACCS) molecular fingerprint, was at least 0.6.
(ii) Therapeutic similarity: the drug pair belongs to the same level 4 ATC subgroup of the WHO ATC classification system, but excluding nonspecific subgroups, which start with the word ‘Other’.
(iii) Difference in hepatotoxicity: one drug of the pair was classified as vNo-DILI-Concern, and the other as vMost-DILI-Concern in DILIrank 2.0.

DILIrank 2.0 contains new drugs added to all 14 first level WHO ATC classification categories. Some drugs have multiple ATC codes and are classified in all relevant therapeutic categories. The relative sizes of most ATC categories are comparable between DILIrank 1.0 and DILIrank 2.0, with one main exception, class L (antineoplastic and immunomodulating agents), which increased from 144 drugs (12%) to 244 drugs (16%) (Figure 2b). The substantial increase reflects the approval by the FDA of 100 cancer-related drugs over the decade.

Based on the second level of the WHO ATC classification system (ATC level 2), which describes the pharmacological or therapeutic action of a drug, 78 antineoplastic agents (L01) and 17 immunosuppressants (L04) were added in DILIrank 2.0. The names of all ATC level 2 subgroups for the DILIrank 2.0 drugs are listed in Table S2 in the supplementary material online. Most of the newly added drugs that were verified as causing DILI (classified as vLess- or vMost-DILI-Concern) belong to these two classes. Distribution of the drugs in DILIrank 2.0 by their therapeutic action and DILI classification reveals that the L01 subgroup has the highest number of DILI-positive (vLess- and vMost-DILI-Concern) drugs (Figure 3). Subgroups enriched in DILI-positive drugs are marked with an

TABLE 1

Differences between DILIrank 1.0 and DILIrank 2.0.

	DILIrank 1.0[a]	DILIrank 2.0[a]	Increase[b]
Total number of drugs	1036	1336	300 (29%)
Number of drugs per DILI group (%)			
vNo-DILI-Concern	312 (30%)	414 (31%)	102 (33%)
Ambiguous-DILI-Concern	254 (25%)	354 (26%)	100 (39%)
vLess-DILI-Concern	278 (27%)	351 (26%)	73 (26%)
vMost-DILI-Concern	192 (19%)	217 (16%)	25 (13%)

[a] Percentage computed based on all groups in database version.
[b] Percentage computed based on each group increase across both database versions.

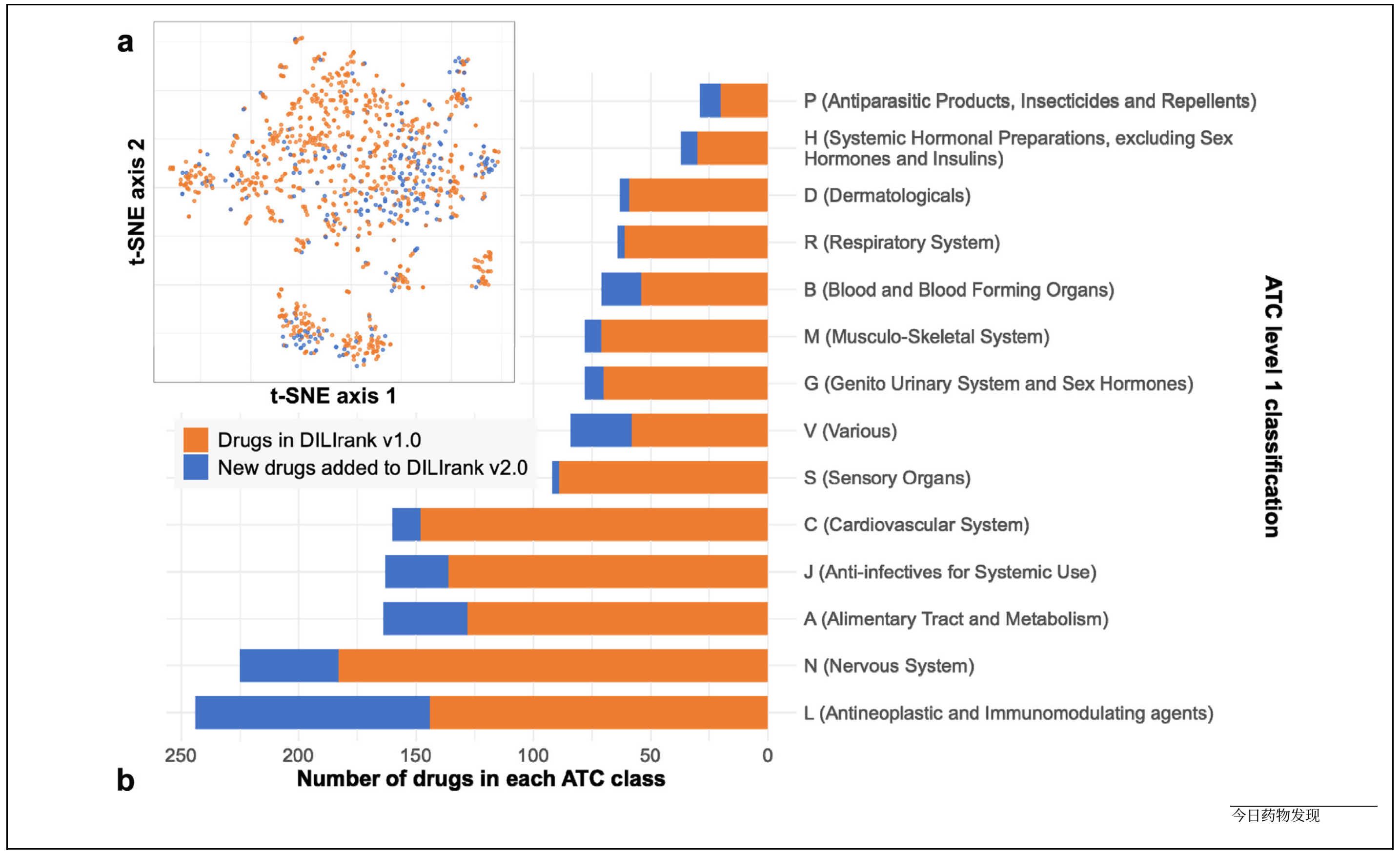

图 2 DILIrank 1.0 与 2.0 的差异：（a）药物的化学空间覆盖，投影在 t-SNE 图上；（b）DILIrank 中按主要解剖 ATC 类别的药物分布（没有 ATC 代码的药物不包括在图中）。

图 3 中的星号。如果富集因子 (EF) 大于或等于 2，则认为一个类别被富集。EF 的计算公式为 P_i N_i P_i N_i，其中 P_i 和 N_i 分别代表每个类别中 DILI 阳性和 DILI 阴性药物（^{v}无 DILI 关注）的数量。

在过去的十年里，科学界对蛋白激酶抑制剂引起的肝毒性产生了显著的兴趣，并报告了几起DILI病例。[(p29)]，（p30），（p31） DILIrank 2.0也反映了这一趋势，因为新添加的蛋白激酶抑制剂中，只有一种被归类为^{v}无DILI关注，而有18种符合我们的验证标准并被归类为 ^{v}低DILI关注或 ^{v}高DILI关注。大多数药物（36种）被归类为模糊DILI关注，因为尽管其药品说明书中有肝毒性警告，但这些药物的病例报告未达到我们的验证标准。

引入FDA监管项目后DILI的特征变化

FDA 的监管政策在确保药物安全方面至关重要，通过执行严格的上市前评估、上市后监测以及风险缓解策略来检测和管理不良药物反应，包括药物性肝损伤（DILI）。DILIrank 2.0 包含了 1939 年至 2021 年 FDA 批准的药物，为评估 FDA 项目引入后药物 DILI 风险概况的趋势提供了前所未有的机会。在本研究中，我们特别考察了三个关键监管项目对获批药物 DILI 责任的影响：(i) FAERS （ FDA 不良事件报告系统），成立于 1969 年；(ii) 加速审批程序，于 1992 年引入；以及 (iii) 针对 DILI 的指导原则，发布于 2009 年。本分析基于 DILIrank 2.0 中的 1265 种药物，这些药物的 FDA 批准年份都可以确定。

我们研究的第一个监管里程碑是1969年FAERS的建立，它引入了一种自发报告药物相关不良事件和用药错误的机制。

在FAERS之前，一系列备受关注的药物不良事件引起了公众对药物安全的担忧。其中最著名的案例之一是沙利度胺，该药在1962年被发现会导致欧洲新生儿严重的先天性异常。[(p33)] 作为回应，美国国会通过了《凯福弗-哈里斯药物修正案》，对《联邦食品、药品和化妆品法》（FDC法）进行了加强监管，以确保药物的疗效和安全。[(p34)] 如图4所示，DILIrank 2.0 包含1939年至1968年（FAERS之前）批准的218种药物，以及1969年至1991年（FAERS之后）批准的309种药物。在此期间， ^{v}最关注和 ^{v}无DILI关注药物的比例从14%下降到

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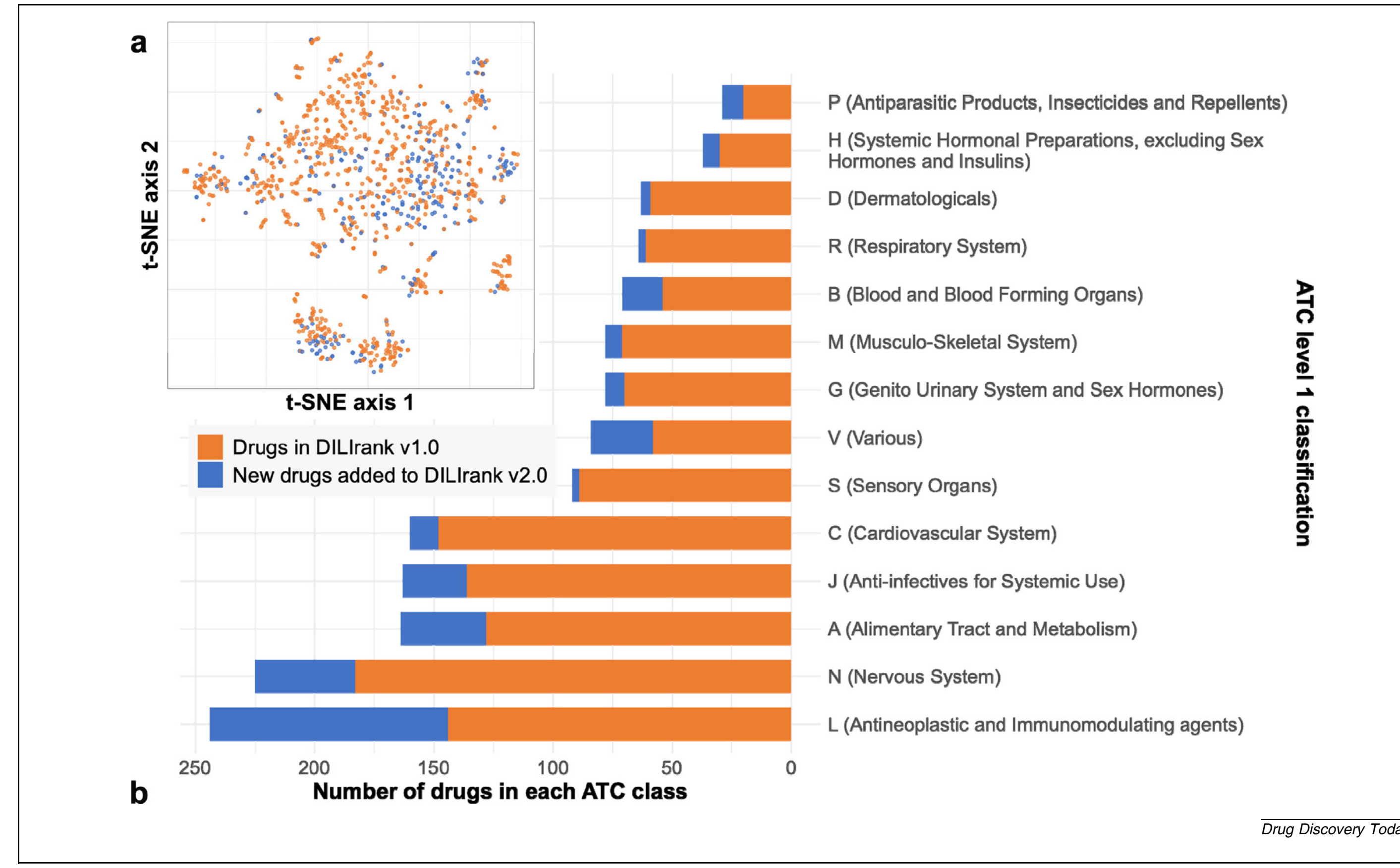

FIGURE 2
Differences between DILIrank 1.0 and 2.0: **(a)** Chemical space coverage of drugs, projected on a t-SNE plot; **(b)**. Distribution of drugs by the main anatomical ATC classes in DILIrank (drugs with no ATC code are excluded from the plot).

asterisk in Figure 3. A class is considered enriched if the enrichment factor (EF) is greater than or equal to 2. EF is calculated as $(P_i/N_i)/(\sum P_i/\sum N_i)$, where P_i and N_i represent the number of DILI-positive and DILI-negative drugs (^{v}No-DILI-Concern) in each class, respectively.

Over the past decade, there has been significant scientific interest in protein kinase inhibitor-induced hepatotoxicity, with several DILI cases reported.[(p29),(p30),(p31)] DILIrank 2.0 also reflects this trend, as only one of the newly added protein kinase inhibitors is classified as ^{v}No-DILI-Concern, whereas 18 met our verification criteria and were classified as ^{v}Less- or ^{v}Most-DILI-Concern. Most of the drugs (36) are classified as Ambiguous DILI-Concern, because despite the hepatotoxicity warning in their drug labels, the case reports for these drugs did not meet our verification criteria.

DILI profile changes following introduction of FDA regulatory programs

The regulatory policies of the FDA are crucial in ensuring drug safety by enforcing rigorous premarket evaluations, post-marketing surveillance, and risk mitigation strategies to detect and manage adverse drug reactions, including DILI. DILIrank 2.0 contains drugs approved by the FDA from 1939 to 2021, offering an unprecedented opportunity to assess trends in the drug DILI risk profile following the introduction of FDA programs. In this study, we specifically examined the influence of three key regulatory programs on the DILI liability of approved drugs: (i) FAERS (FDA Adverse Event Reporting System), established in 1969; (ii) expedited approval programs, introduced in 1992; and (iii) the DILI-specific guidance, issued in 2009. The analysis is based on 1265 drugs in DILIrank 2.0 for which we could identify the year of FDA drug approval.

The first regulatory milestone we examined was the establishment of FAERS in 1969, which introduced a mechanism for the spontaneous reporting of drug-related adverse events and medication errors.[(p32)] Before FAERS, a series of high-profile drug-induced adverse events heightened public concern over drug safety. One of the most notable cases was thalidomide, which was found in 1962 to cause severe congenital abnormalities in newborns across Europe.[(p33)] In response, the US Congress passed the Kefauver–Harris Drug Amendments to the Federal Food, Drug, and Cosmetic (FDC) Act, strengthening regulations to ensure both drug efficacy and safety.[(p34)] As illustrated in Figure 4, DILIrank 2.0 contains 218 drugs approved between 1939 and 1968 (pre-FAERS) and 309 drugs approved between 1969 and 1991 (post-FAERS). Over this period, the proportion of ^{v}Most- and ^{v}No-DILI-concern drugs decreased from 14% to

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表 2

具有与DILI相关的活性悬崖的药物对。

S/N	LTKBID	药品名称	DILI-关注	田本相似度	治疗类别
1	LT02444 LT01600	恩扎卢胺 尼鲁他胺	^{v}不- ^{v}大多数-	0.63	抗雄激素
2	LT03745 LT00034	利福霉素钠 利福平	^{v}不- ^{v}大多数-	0.64	抗生素
3	LT03753 LT01222	盐酸沙雷霉素 泰格环素	^{v}不- ^{v}大多数-	0.77	四环素类
4	LT03753 LT00416	盐酸沙雷霉素 盐酸米诺环素	^{v}不- ^{v}大多数-	0.79	四环素类
5	LT03563 LT00380	沃克洛司钠 环孢素	^{v}不- ^{v}大多数-	0.98	钙调神经磷酸酶抑制剂

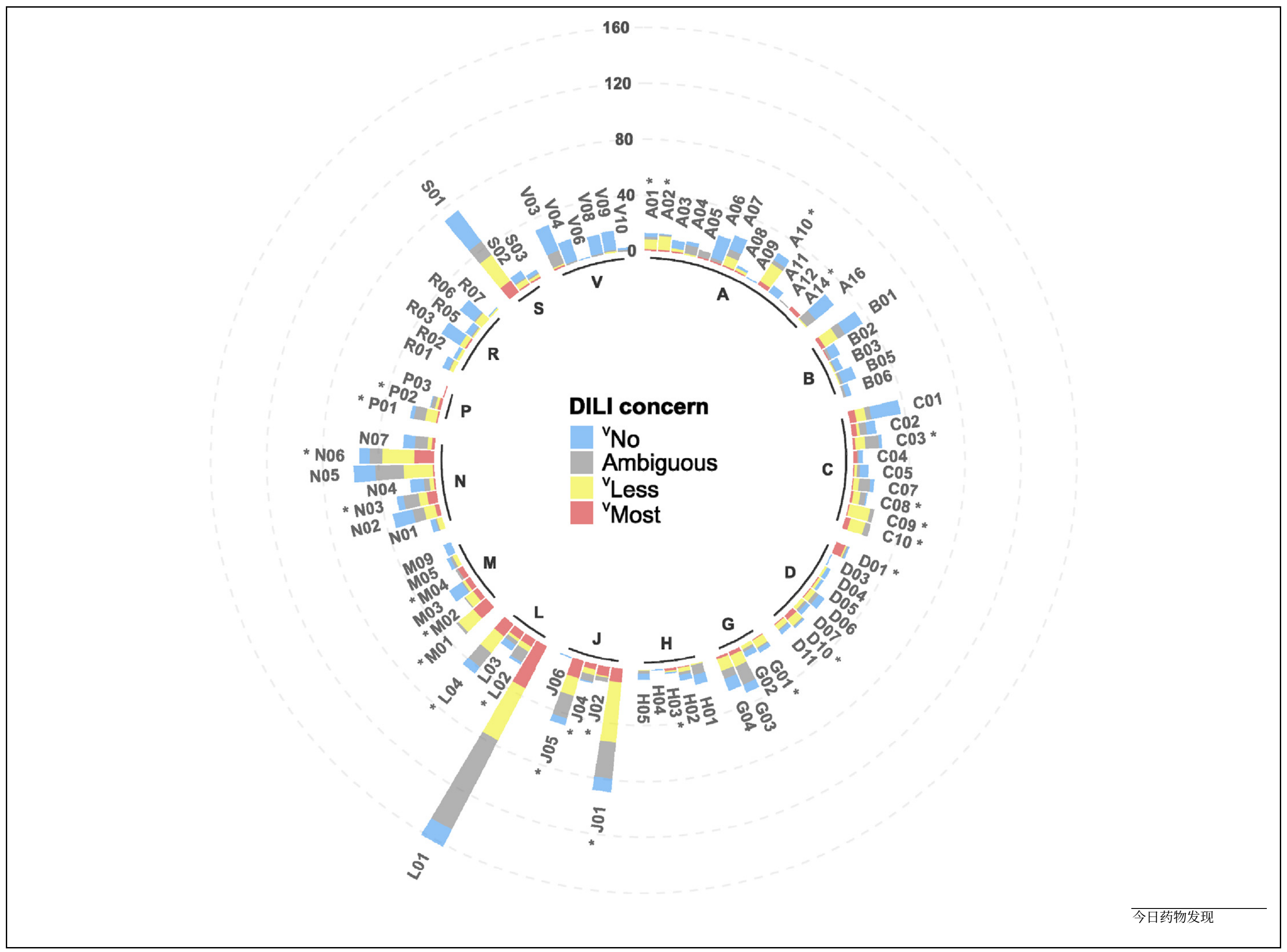

图3 DILIrank 2.0 按 WHO ATC 二级类别（药理学或治疗学亚组）和 DILI 分类的分布。每个条的高度表示该类别中药物的数量。所有 ATC 二级亚组的名称列于表 S2。

TABLE 2

Drug pairs with DILI-related activity cliffs.

S/N	LTKBID	Drug name	DILI-Concern	Tanimoto similarity	Therapeutic category
1	LT02444 LT01600	Enzalutamide Nilutamide	^{v}No- ^{v}Most-	0.63	Anti-androgens
2	LT03745 LT00034	Rifamycin sodium Rifampin	^{v}No- ^{v}Most-	0.64	Antibiotics
3	LT03753 LT01222	Sarecycline hydrochloride Tigecycline	^{v}No- ^{v}Most-	0.77	Tetracyclines
4	LT03753 LT00416	Sarecycline hydrochloride Minocycline hydrochloride	^{v}No- ^{v}Most-	0.79	Tetracyclines
5	LT03563 LT00380	Voclosporin Cyclosporine	^{v}No- ^{v}Most-	0.98	Calcineurin inhibitors

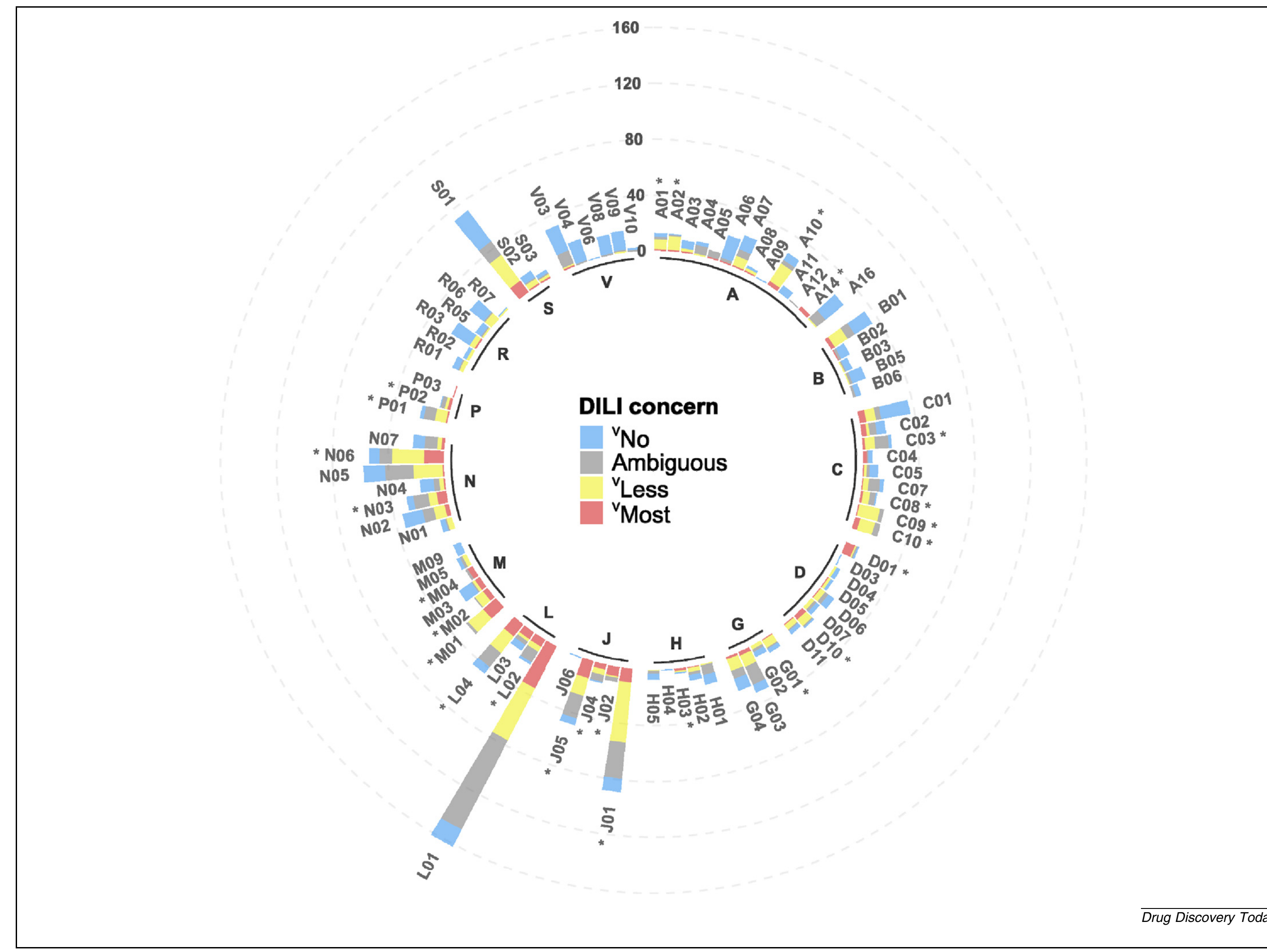

FIGURE 3

Distribution of DILIrank 2.0 by the WHO ATC level 2 class (pharmacological or therapeutic subgroup) and DILI classification. Each bar height represents the number of drugs in each class. The names of all ATC level 2 subgroups are listed in Table S2.

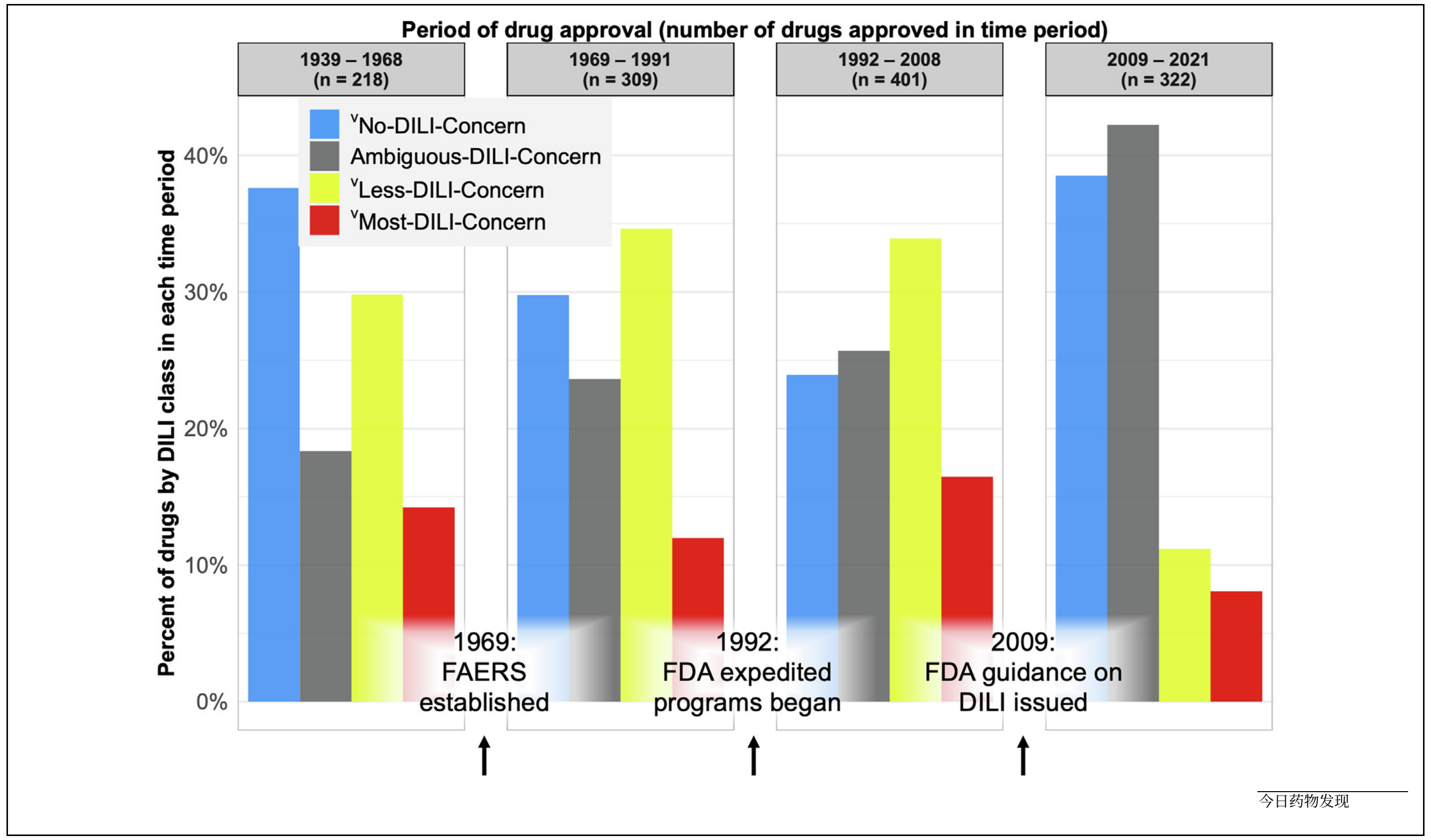

图4 DILIrank 2.0 中按 DILI 分类和批准年份分类的药物。

在FAERS系统实施前后，这一比例分别从12%和38%降至30%。与此同时，低度和模糊DILI关注药物的比例略有增加。根据这一时期所有四组的变化，很难确定FAERS系统对FDA批准药物DILI风险特征的具体影响。然而，值得注意的是，自1997年经过重大修订以来，FAERS数据库已成为此类报告的最大存储库，其数据已被多次挖掘以建立药物与其报告的不良事件之间的关联。

在1980年代晚期，FDA正式表示其有兴趣“加快有前景的新疗法的可用性”，以加速针对严重未满足医疗需求的药物的开发和批准。FDA采用了四个项目，这些项目在不同时间启动，以加快其审查和批准过程：优先审评（1992）、加速批准（1992）、快速通道（1998）和突破性疗法批准（2012）。根据对1987年至2023年间FDA批准的药物的多项研究，快速通道和优先审评是这四种FDA加速项目中使用最频繁的两种。

p38),(p39),(p40),(p41),(p42) 然而，人们担心加快审批流程可能与上市后药物安全问题的发生率更高有关。[p43],[p44] 如图4所示，与1969年至1991年批准的药物相比，我们观察到1992年至2008年批准的药物中 ᵛ最受DILI关注的药物比例更高（16%）。同时，ᵛ无DILI关注药物的比例从30%（1969–1991年）下降到24%（1992–2008年）。这些发现表明，通过加快审批程序批准的药物可能具有更高的肝损伤风险，强调评估加强上市后监测以监控和减轻潜在安全问题的需求。同样值得注意的是，加速审批项目通常用于那些在面对危及生命和严重致残疾病的治疗时“愿意接受更大风险和副作用”的药物。与此一致，DILIrank 2.0显示，通过FDA加速审批项目获批并被归类为“最高DILI关注”（Most-DILI-Concern, 见在线补充材料表S3）的药物，大多数是抗肿瘤药（ATC代码：L01）和全身用抗病毒药（ATC代码：J05）。

2009年，FDA发布了关于进行DILI上市前临床评估的指导意见。值得注意的是，新批准药物的肝脏安全性特征在随后的几年（2009–2021年）显著改善。在此期间，

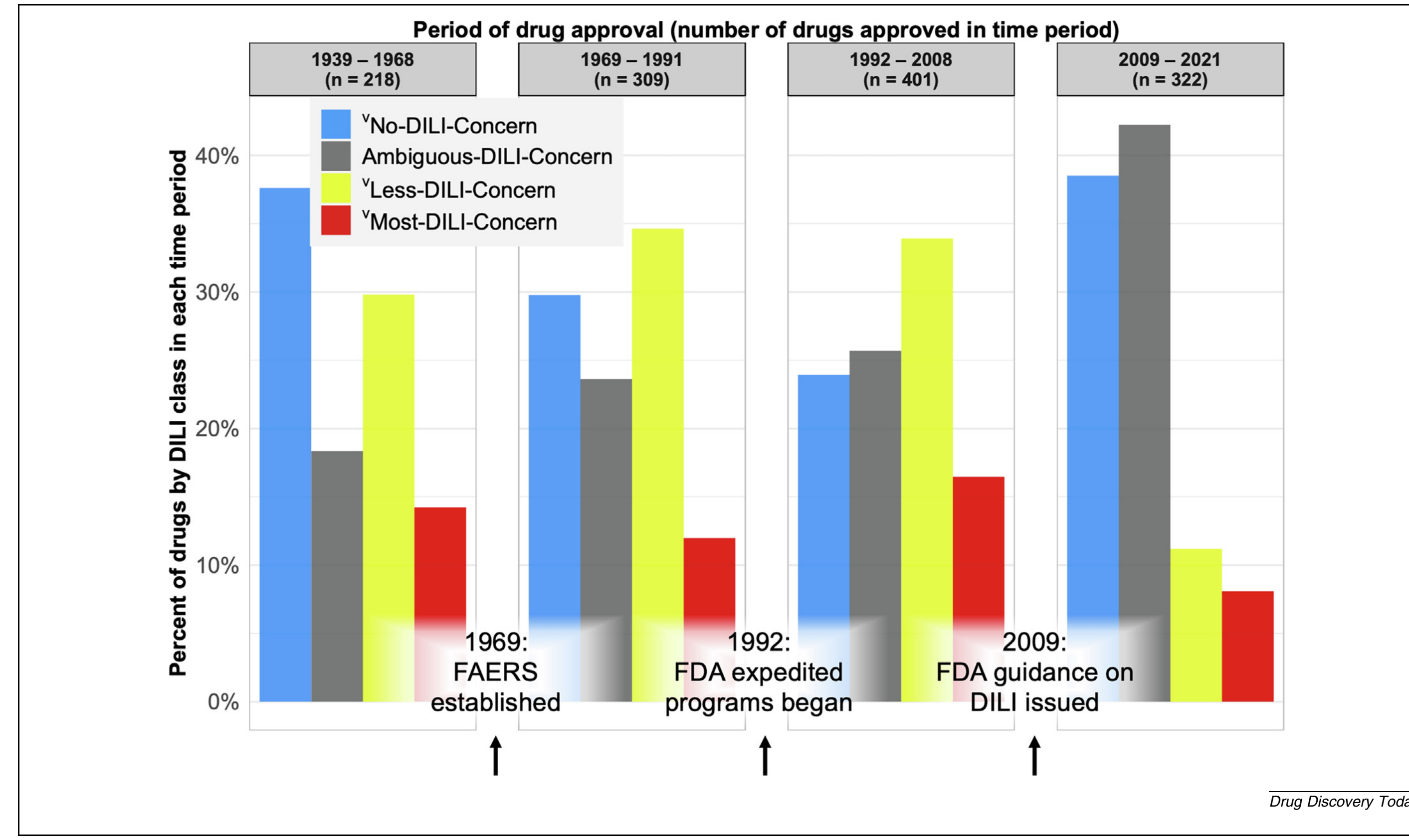

FIGURE 4
Drugs in DILIrank 2.0 by DILI classification and year of approval.

12% and from 38% to 30%, respectively, during the pre-FAERS to the post-FAERS era. Meanwhile, the proportion of ᵛLess- and Ambiguous-DILI-Concern drugs slightly increased. Based on the changes in all four groups during this period, it is difficult to ascertain the specific influence of the FAERS system on the DILI risk profile of FDA approved drugs. Nevertheless, it is worth noting that since its major revision in 1997, the FAERS database has become the largest repository of such reports, and its data have been mined several times to establish links between drugs and their reported adverse events.[p35]

In the late 1980s, the FDA officially stated its interest in 'expediting the availability of promising new therapies' to accelerate the development and approval of drugs addressing serious unmet medical needs.[p36] The FDA employed four programs, initiated at different times to expedite its review and approval process: Priority Review (1992), Accelerated Approval (1992), Fast Track (1998), and Breakthrough Therapy Approval (2012).[p37] According to multiple investigations into drugs approved by the FDA between 1987 and 2023, Fast Track and Priority Review are the two most used of the four FDA expedited programs.[p37],[p38],[p39],[p40],[p41],[p42] However, concerns have been raised that expedited approval processes might be associated with a higher likelihood of post-marketing drug safety issues.[p43],[p44] As shown in Figure 4, compared with drugs approved between 1969 and 1991, we observed a higher proportion of ᵛMost-DILI-Concern drugs among those approved during 1992–2008 (16%). Simultaneously, the proportion of ᵛNo-DILI-Concern drugs declined from 30% (1969–1991) to 24% (1992–2008). These findings suggest that drugs approved through expedited programs might carry a higher risk of DILI, emphasizing the need for enhanced post-marketing surveillance to monitor and mitigate potential safety concerns. It is also worth noting that expedited programs are generally employed for drugs to be used in cases where there is a willingness to 'accept greater risks and side effects from treatment of life-threatening and severely debilitating disease'.[p45] In line with this, DILIrank 2.0 shows that the majority of the drugs that were approved through an FDA expedited program and are classified as ᵛMost-DILI-Concern (see Table S3 in supplementary material online) are antineoplastic agents (ATC code: L01) and antivirals for systemic use (ATC code: J05).

In 2009, the FDA issued guidance on conducting premarketing clinical evaluation of DILI.[p46] Notably, the liver safety profile of newly approved drugs has significantly improved in subsequent years (2009–2021). During this period, the ᵛ无肝损伤担忧药物的比例增加至39%；而DILI阳性药物以及 ᵛ较少和ᵛ高度肝损伤担忧药物的比例则分别下降至历史最低的11%和8%。这些发现表明，在DILI指南引入后，FDA批准药物相关的DILI风险呈现出积极的下降趋势。然而，需要注意的是，这一时期模糊肝损伤担忧药物的比例也达到了最高值（42%）。这可能反映了在临床试验和药物标签中，对肝毒性问题的报告意识和谨慎性提高。由于上市后监测需要足够的时间来发现后期出现的安全信号，未来仍将持续需要评估和更新这些药物的DILI风险。

结束语

DILIrank 2.0 是一个更新和扩展的数据集，包含对 1939 年至 2021 年 FDA 批准的 1336 种药物的 DILI 分类，分为四类 DILI 关注类别。DILIrank 2.0 提供了一个宝贵的资源，以支持对 DILI 的进一步研究，特别是那些使用 NAMs（包括人工）的方法。基于人工智能（AI）的方法。根据这一更新，我们观察到在FDA监管项目引入后的八十年间获批药物的肝脏安全性特征发生了变化。随着药物开发工作的推进，将需要继续为新药标注其可能引起药物性肝损伤（DILI）的潜力，并在出现新证据时重新评估已有药物的标注。

利益声明

没有需要声明的利益冲突。

CRediT 作者贡献声明

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本文反映了作者的观点，并不一定代表美国食品药品监督管理局的观点。任何对商业产品的提及仅用于说明，并不表示批准、认可或推荐。

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附录 A. 补充材料

本文的补充材料可在网上找到，网址为 https://doi.org/10.1016/j.drudis.2025.104485。

数据可用性

DILIrank 2.0 数据集可在线获取，网址为 https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-rank-DILIrank-dataset。

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proportion of ᵛNo-DILI-Concern drugs increased to 39%; whereas the proportions of DILI-positive drugs and ᵛLess- and ᵛMost-DILI-Concern drugs dropped to all-time lows of 11% and 8%, respectively. These findings suggest that following the introduction of the DILI guidance there has been a positive trend in mitigating DILI risk associated with FDA-approved drugs. However, it is important to note that this period also records the highest proportion of Ambiguous DILI-Concern drugs (42%). This might reflect raised awareness and caution in reporting hepatotoxicity concerns during clinical trials, and in drug labels. Because post-marketing surveillance requires sufficient time to detect late-emerging safety signals, there will be a continuous need to assess and update the DILI risk of these drugs in the future.

Concluding remarks

DILIrank 2.0 is an updated and expanded dataset with DILI classification for 1336 drugs approved by the FDA from 1939 to 2021 under four DILI-concern categories. DILIrank 2.0 offers a valuable resource to enable further studies on DILI, particularly those using NAMs including artificial intelligence (AI)-based approaches. Based on this update, we observed changes in liver safety profiles of drugs approved over a period of eight decades following the introduction of FDA regulatory programs. As drug development efforts advance, there will be a need to continue annotating new drugs for their potential to cause DILI and to reassess the annotations of former drugs as new evidence emerges.

Declarations of interest

There are no conflicts of interest to declare.

CRediT authorship contribution statement

AyoOluwa O. Olubamiwa: Data curation, Formal analysis, Methodology, Writing – original draft; **Yanyan Qu:** Data curation, Formal analysis, Writing – review and editing; **Skylar Connor:** Formal analysis, Writing – review and editing; **Weida Tong:** Conceptualization, Formal analysis, Writing – review and editing; **Dongying Li:** Data curation, Formal analysis, Project administration, Writing – review and editing; **Minjun Chen:** Conceptualization, Data curation, Formal analysis, Methodology, Writing – original draft, Writing – review and editing.

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Appendix A. Supplementary material

Supplementary material to this article can be found online at https://doi.org/10.1016/j.drudis.2025.104485.

Data availability

The DILIrank 2.0 dataset is available online at https://www.fda.gov/science-research/liver-toxicity-knowledge-base-ltkb/drug-induced-liver-injury-rank-DILIrank-dataset.

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